

Section 2.1

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June 5, 2024

Exercise 1. Let $f : X \rightarrow \overline{\mathbb{R}}$ and $Y = f^{-1}(\mathbb{R})$. Then f is measurable iff $f^{-1}(\{-\infty\}) \in \mathcal{M}$, $f^{-1}(\{\infty\}) \in \mathcal{M}$, and f is measurable on Y .

Proof. First, assume that $f^{-1}(\{-\infty\}), f^{-1}(\{\infty\}) \in \mathcal{M}$ and f is measurable on Y . Then given $a \in \mathbb{R}$,

$$f^{-1}((a, \infty]) = f^{-1}((a, \infty)) \cup f^{-1}(\{\infty\})$$

is measurable because the right hand side is a union of measurable sets since $(a, \infty) \subset \mathbb{R}$.

On the other hand, if f is measurable, then clearly since $\{-\infty\}, \{\infty\} \in \mathcal{B}_{\overline{\mathbb{R}}}$, $f^{-1}(\{-\infty\}), f^{-1}(\{\infty\}) \in \mathcal{M}$. Similarly, if $a \in \mathbb{R}$, then $(a, \infty) \in \mathcal{B}_{\overline{\mathbb{R}}}$ so $f^{-1}((a, \infty)) \cap Y = f^{-1}((a, \infty)) \in \mathcal{M}$. $\square$

Exercise 2. Suppose $f, g : X \rightarrow \overline{\mathbb{R}}$ are measurable.

- a. fg is measurable (where $0 \cdot (\pm\infty) = 0$).
- b. Fix $a \in \overline{\mathbb{R}}$ and define $h(x) = a$ if $f(x) = -g(x) = \pm\infty$ and $h(x) = f(x) + g(x)$ otherwise. Then h is measurable.

Proof. a. First define $\phi : X \rightarrow \overline{\mathbb{R}}^2$ by $\phi(x) = (f(x), g(x))$. Proposition 2.4 shows that ϕ is $(\mathcal{M}, \mathcal{B}_{\overline{\mathbb{R}}^2})$ -measurable. Then define $\psi : \overline{\mathbb{R}}^2 \rightarrow \overline{\mathbb{R}}$ by

$$\begin{aligned} \psi((x, y)) &= xy & (x, y \in \mathbb{R}) \\ 0 \cdot (\pm\infty) &= 0 \\ a \cdot (\pm\infty) &= \pm\infty & (a > 0) \\ a \cdot (\pm\infty) &= \mp\infty & (a < 0) \end{aligned}$$

We'll use Exercise 1, first showing that ψ is measurable on $\psi^{-1}(\mathbb{R})$. Note that

$$\psi^{-1}(\mathbb{R})$$

This is simple since we consider $\psi|_{\psi^{-1}(\mathbb{R})}$ with respect to the algebra

$$\begin{aligned} \{E \cap \mathbb{R}^2 : E \in \mathcal{B}_{\overline{\mathbb{R}}^2}\} &= \{E \cap \mathbb{R}^2 : E \in \mathcal{B}_{\overline{\mathbb{R}}} \times \mathcal{B}_{\overline{\mathbb{R}}}\} \\ &= \mathcal{B}_{\mathbb{R}^2}, \end{aligned}$$

so $\psi|_{\psi^{-1}(\mathbb{R})}$ is just multiplication on $\mathbb{R}$, which is continuous.

We now consider that

$$\begin{aligned}\psi^{-1}(\{\infty\}) &= (\overline{\mathbb{R}}^+ \times \{\infty\}) \cup (\{\infty\} \times \overline{\mathbb{R}}^+) \\ &\quad \cup (\{-\infty\} \times \overline{\mathbb{R}}^-) \cup (\overline{\mathbb{R}}^- \times \{-\infty\})\end{aligned}$$

and, similarly,

$$\begin{aligned}\psi^{-1}(\{-\infty\}) &= (\overline{\mathbb{R}}^- \times \{-\infty\}) \cup (\{-\infty\} \times \overline{\mathbb{R}}^-) \\ &\quad \cup (\{\infty\} \times \overline{\mathbb{R}}^+) \cup (\overline{\mathbb{R}}^+ \times \{\infty\}).\end{aligned}$$

Both are clearly measurable since they are the simple union of measurable sets, so Exercise 1 leads us to conclude that ψ is $(\mathcal{B}_{\mathbb{R}^2}, \mathcal{B}_{\overline{\mathbb{R}}})$ -measurable.

Thus, since $fg = \psi \circ \phi$, fg is $(\mathcal{M}, \mathcal{B}_{\overline{\mathbb{R}}})$ -measurable.

b.

□